

Please check the examination details below before entering your candidate information

Candidate surname						Other names					
Centre Number						Learner Registration Number					

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Wednesday 22 May 2024

Afternoon (Time: 40 minutes)

Paper reference **31617H/1P**

Applied Science/Forensic and Criminal Investigation

UNIT 1: Principles and Applications of Science I

Physics

SECTION C: WAVES IN COMMUNICATION

You must have:
a calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each:
 - Section A: Structures and Functions of Cells and Tissues (Biology)
 - Section B: Periodicity and Properties of Elements (Chemistry)
 - Section C: Waves in Communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*
- The formulae sheet can be found at the back of this paper.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

- 1 (a) Figure 1 shows a transverse wave travelling along a string.

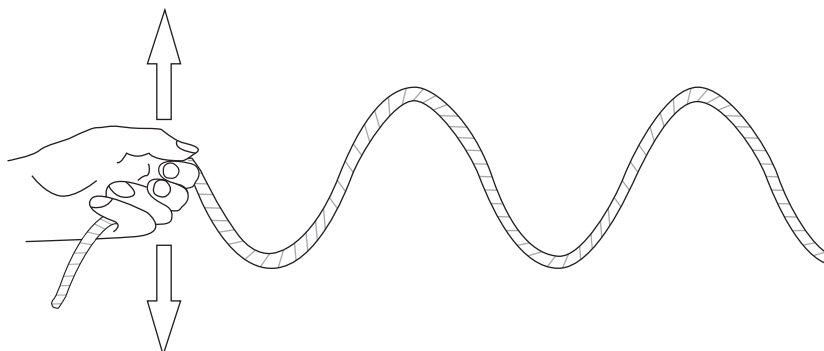


Figure 1

Figure 2 represents the displacement of the string with distance along the string at one time.

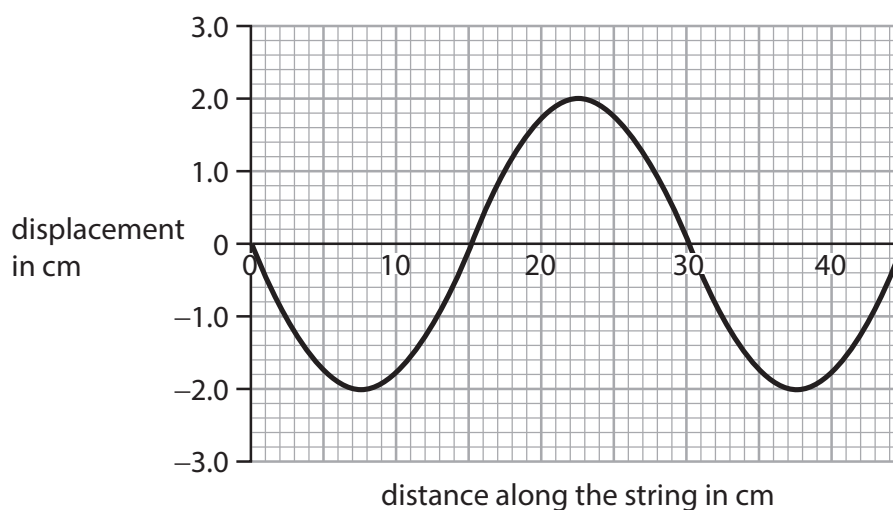


Figure 2

- (i) Identify the amplitude of the wave in Figure 2.

(1)

amplitude = cm

- (ii) Identify the wavelength of the wave in Figure 2.

(1)

wavelength = cm



Figure 3 shows a standing wave being produced on a long spring.

The wave is produced at point P.

The wave is reflected at point Q.

Point Q is stationary.

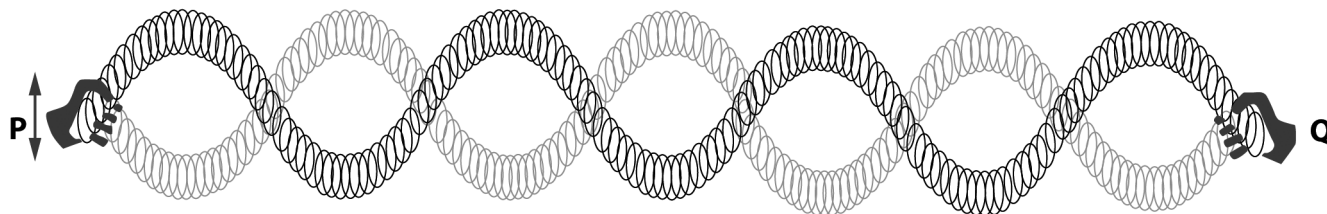


Figure 3

(b) Which **one** of these is the phase change of the spring at point Q?

(1)

- ☐ A 45°
- ☐ B 90°
- ☐ C 180°
- ☐ D 360°

A wave travels 3.5 metres in 2.2 seconds.

(c) Calculate the speed of the wave.

Use the equation wave speed = $\frac{\text{distance}}{\text{time}}$

(2)

Show your working.

wave speed = ms^{-1}

(Total for Question 1 = 5 marks)



2 Sound waves are longitudinal waves.

(a) Complete the following sentence using words from the box:

(2)

compression	conduction	rarefaction
reflection	refraction	

Longitudinal waves in the air can be described as a series of

..... and

A bat finds its prey using echolocation.

This means that the bat uses sound waves to hunt for its prey.

Figure 4 shows a bat and its prey.



(Source: ©PAL)

Figure 4

(b) Describe how the bat can find the position of its prey.

(3)

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(Total for Question 2 = 5 marks)



- 3 (a) Figure 5 shows the emission spectra of five elements that may be in stars.

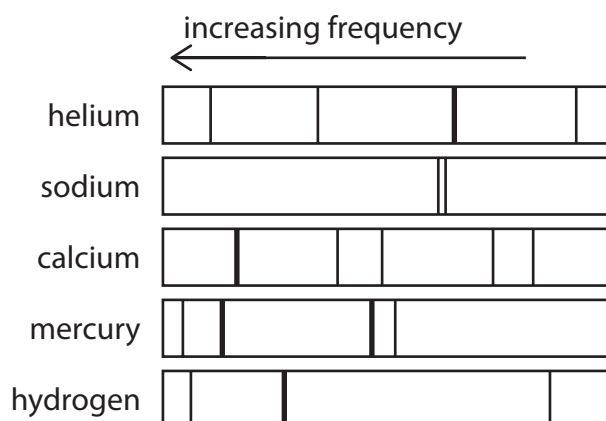


Figure 5

Figure 6 shows the emission spectra emitted by four stars.

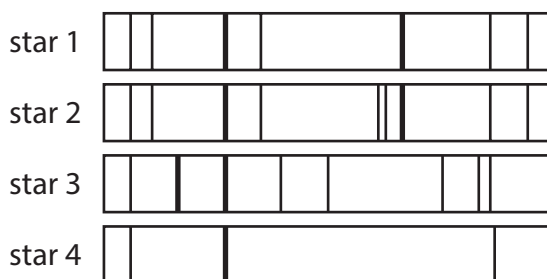


Figure 6

- (i) Which **one** of these stars contains the element sodium?

Use the information in Figure 5 and Figure 6.

(1)

- ☐ **A** star 1
- ☐ **B** star 2
- ☐ **C** star 3
- ☐ **D** star 4

- (ii) Explain which of the elements shown in Figure 5 is present in the emission spectra of **all** the stars shown in Figure 6.

(2)

.....

.....

.....

.....



The atoms of an element are heated.

This causes electrons to move to higher energy levels.

Electrons are not stable when they are in higher energy levels.

- (b) Explain how changes in the energy of the electrons make the emission spectrum of each element different.

(3)

(Total for Question 3 = 6 marks)



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4 Total internal reflection of light only happens in specific conditions.

(a) Which **one** of these conditions would allow total internal reflection?

(1)

		light travels	incident angle for the light
☐	A	from air into glass	is greater than the critical angle
☐	B	from air into glass	is less than the critical angle
☐	C	from glass into air	is greater than the critical angle
☐	D	from glass into air	is less than the critical angle



Figure 7 represents light travelling from air into glass and being refracted.

The angle of incidence (i) in air is 40° .

The angle of refraction (r) in glass is 25° .

The velocity of light in air (c) is $3.0 \times 10^8 \text{ m s}^{-1}$.

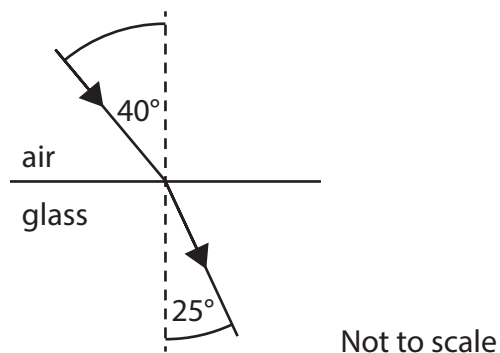


Figure 7

(b) Calculate the velocity of light (v) in the glass.

Use the equation $\frac{c}{v} = \frac{\sin i}{\sin r}$

(4)

Show your working.

velocity of light in glass = ms^{-1}

(Total for Question 4 = 5 marks)



5 Bluetooth® and Wi-Fi use overlapping frequencies to transmit signals.

(a) Give a reason why the signals from Bluetooth® and Wi-Fi do not interfere.

(1)

Bluetooth® and infrared are both used to transmit signals for short distances.

(b) Give **two** differences between Bluetooth® and infrared transmissions.

(2)

1

2

A smartphone is not always able to access the internet, make or receive calls and send or receive text messages.

(c) Discuss the reasons why transmission and reception of signals to and from smartphones can be disrupted.

(6)



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(Total for Question 5 = 9 marks)

TOTAL FOR PAPER = 30 MARKS
TOTAL FOR EXAMINATION = 90 MARKS



Formulae sheet

Wave speed

$$v = f\lambda$$

Speed of a transverse wave on a string

$$v = \sqrt{\frac{T}{\mu}}$$

Refractive index

$$n = \frac{c}{v} = \frac{\sin i}{\sin r}$$

Critical angle

$$\sin C = \frac{1}{n}$$

Inverse square law in relation to the intensity of a wave

$$I = \frac{k}{r^2}$$

$$\frac{I_1}{I_2} = \frac{(D_2)^2}{(D_1)^2}$$

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